



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE

4003/1

PAPER 1 Multiple Choice

NOVEMBER 2019 SESSION

1 hour

Additional materials:
Multiple Choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended.)
Calculator (Optional)

INSTRUCTIONS TO CANDIDATES

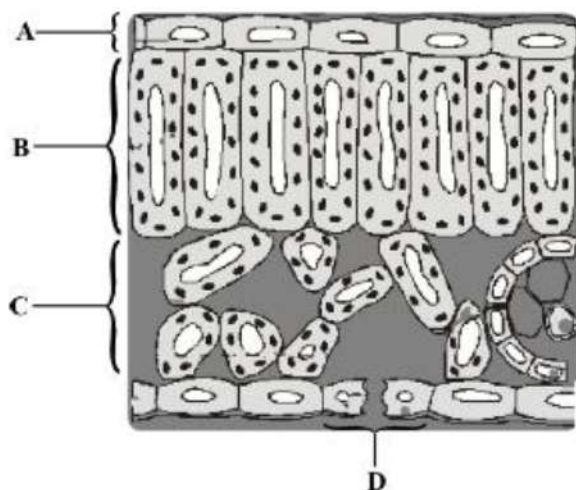
Do **not** open this booklet until you are told to do so.
Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.
Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet provided.

1. The diagram shows the internal structure of a leaf.
In which tissue, **A**, **B**, **C** or **D**, does most photosynthesis take place?

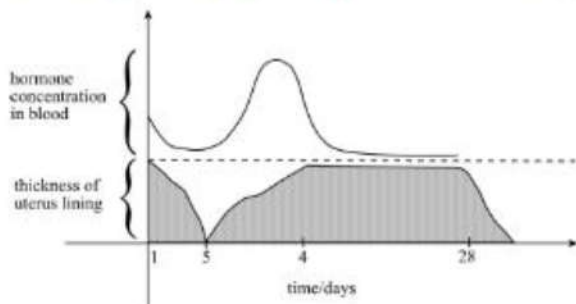


2. Which cell is specialised for contraction?
- A sperm cell
 - B muscle cell
 - C red blood cell
 - D white blood cell
3. Overfeeding causes
- A goitre.
 - B scurvy.
 - C obesity.
 - D anorexia nervosa.

4. Which teeth are used for cutting food?
- A premolars
 - B molars
 - C canines
 - D incisors
5. The equation for respiration is given below:
 $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{P} + \text{energy}$
Product P is
- A lactic acid.
 - B alcohol.
 - C starch.
 - D water.
6. What is the function of the enzyme amylase in digestion?
- A to break down fats to fatty acids and glycerol
 - B to break down proteins to amino acids
 - C to break down starch to sucrose
 - D to break down starch to maltose
7. As compared to the aorta, the vena cava
- A has no valves.
 - B has a wider lumen.
 - C has a wall with more elastic fibres.
 - D carries blood under high pressure.

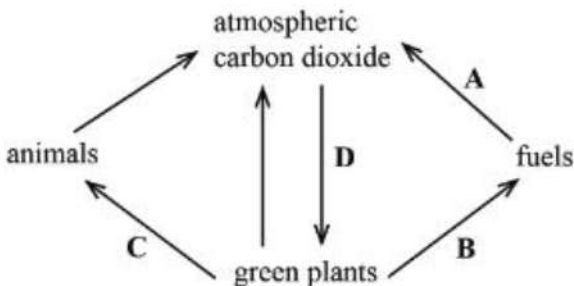
8. The rate of transpiration is low when there is high
- A humidity.
 - B wind speed.
 - C temperature.
 - D light intensity.
9. Wind pollinated flowers have
- A exposed anthers.
 - B enclosed stigma.
 - C coloured petals.
 - D sweet nectar.
10. Which part of the male reproductive system produces sperms?
- A epididymis
 - B sperm duct
 - C urethra
 - D testis

11. The graph shows some of the changes during the 28 day menstrual cycle.



What is the function of the hormone?

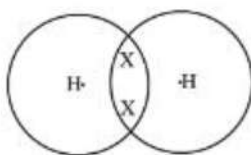
- A causes ovulation
 - B prevents ovulation
 - C rebuilds the uterus lining
 - D maintains the uterus lining
12. The diagram shows the carbon cycle.
- Which process, A, B, C or D, causes global warming?



13. Which statement is true about the blood circulatory system?
- A It is composed of systemic circulation only.
 - B It is composed of pulmonary circulation only.
 - C The blood moves twice into the heart in one cycle.
 - D The left side of the heart receives deoxygenated blood.

14. Which organism causes malaria?
- A virus
 - B fungus
 - C bacterium
 - D plasmodium
15. An element X has the nuclide notation p_qX .
What does p represent?
- A mass number
 - B proton number
 - C neutron number
 - D electron number
16. The electronic configuration of a sodium atom is 2, 8, 1 and that of an oxygen atom is 2, 6.
What is the formula of the compound formed between sodium and oxygen?
- A NaO
 - B Na₂O
 - C NaO₂
 - D NaO₆

17. The diagram shows the type of bonding in a molecule. **H** represents an atom and **X** represents an electron.

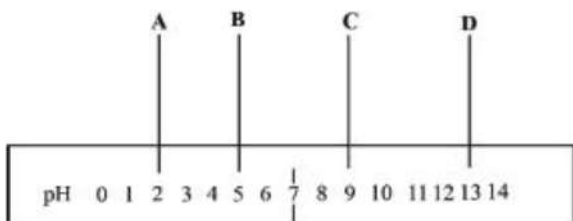


What is the type of bonding shown in the diagram?

- A ionic because electrons are shared
 - B ionic because electrons are transferred
 - C covalent because electrons are shared
 - D covalent because electrons are transferred
18. Which method is used to separate a mixture of dyes?
- A filtration
 - B decanting
 - C evaporation
 - D chromatography
19. Which gas is produced when a metal reacts with an acid?
- A carbon dioxide
 - B hydrogen
 - C oxygen
 - D nitrogen

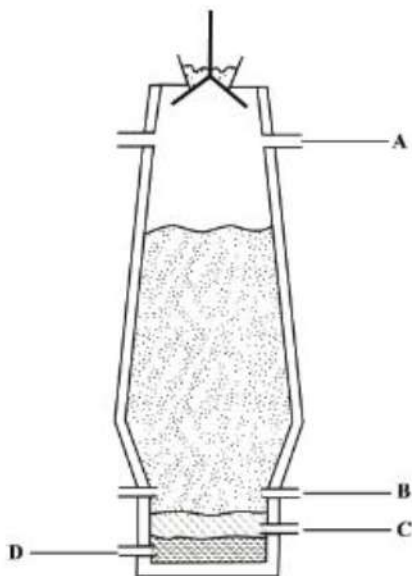
20. Substances **A**, **B**, **C** and **D** have pH values shown in the diagram.

Which substance is a strong acid?



21. The diagram shows the blast furnace.

At which point, **A**, **B**, **C** or **D**, is iron collected?

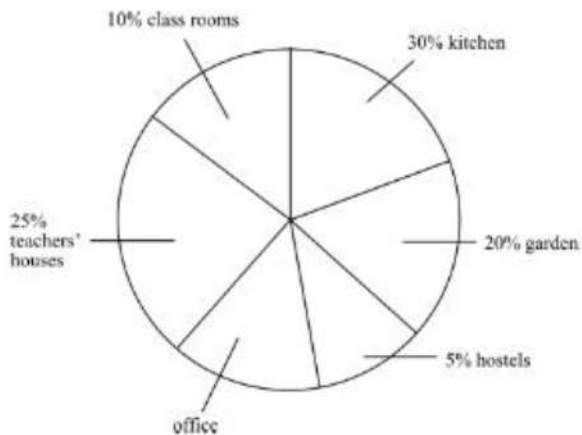


22. Which gas is produced during the electrolysis of molten lead bromide?
- A hydrogen
 - B oxygen
 - C bromine
 - D chlorine
23. Which one is a raw material needed to produce sulphuric acid?
- A oleum
 - B nitrogen
 - C sulphur dioxide
 - D sulphur trioxide
24. What is the correct ratio of nitrogen to hydrogen in the production of ammonia?
- A 1 : 1
 - B 3 : 2
 - C 1 : 3
 - D 2 : 3
25. Ethene is used to make
- A explosives.
 - B fertilisers.
 - C plastics.
 - D soap.
26. Which gas is used in the manufacture of margarine?
- A chlorine
 - B hydrogen
 - C nitrogen
 - D oxygen

27. A hydrocarbon is a molecule which contains atoms of

- A carbon and oxygen.
- B carbon and hydrogen.
- C hydrogen and oxygen.
- D carbon, hydrogen and oxygen.

28. A boarding school uses \$10 000. 00 per month on its running costs.
The pie chart shows the percentages of the costs at the school per month.



How much money is spent in running the office?

- A \$500.00
- B \$900.00
- C \$1 000.00
- D \$2 500.00

29. Which one is a derived unit?

- A metre
- B second
- C newton
- D kilogram

30. The mass of 25 cm^3 of liquid **K** is 50 g.

What is the density of liquid **K**?

- A 0.5 g / cm^3
- B 2.0 g / cm^3
- C 25.0 g / cm^3
- D 1250.0 g / cm^3

31. What is the pressure when a force of 100 N acts on an area of 25 m^2 ?

- A 2500.00 Pa
- B 75.00 Pa
- C 4.00 Pa
- D 0.25 Pa

32. What is the correct formula for calculating pressure in liquids?
[h represents the height of the liquid, ρ represents the density of the liquid and g represents the acceleration due to gravity]

A $P = \frac{h\rho}{g}$

B $P = h\rho g$

C $P = \frac{hg}{\rho}$

D $P = \frac{\rho g}{h}$

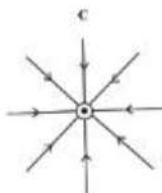
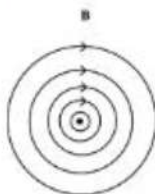
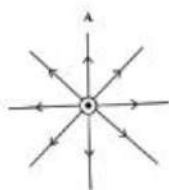
33. Heat from the sun reaches the earth through

- A conduction.
- B convection.
- C radiation.
- D absorption.

34. Why are black pots used on solar cookers?

- A Black is a good reflector of heat.
- B Black is a good absorber of heat.
- C Black is a good insulator of heat.
- D Black focuses heat to a point.

35. Which diagram, A, B, C or D, represents the pattern and the direction of the magnetic field around a straight wire carrying a current out of the page?



Key

● current out of the page

36. Why are electric cables insulated?
- A to prevent loss of electricity
 - B to prevent over heating
 - C to prevent short circuits
 - D to prevent over loading
37. Which statement, about a fuse, is correct?
- A An electric circuit only works if it has a fuse.
 - B A fuse protects the appliance from high current.
 - C A fuse should be connected to the neutral wire in a plug.
 - D An earth wire is needed to prevent the fuse from blowing up.
38. A household uses 250 kWh of electricity per month. The unit cost of electricity is 20 c.
- What is the total cost of the electricity used by the household per month?
- A \$125.00
 - B \$50.00
 - C \$12.50
 - D \$5.00
39. Sound signal in an optic fibre is transmitted as
- A heat energy.
 - B light energy.
 - C sound energy.
 - D electrical energy.
40. Which one is a function of an antenna?
- A to receive signals
 - B to produce the carrier wave
 - C to shorten the range of the signals
 - D to separate the carrier signals and information signals



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

COMBINED SCIENCE

4003/3

PAPER 3 (Practical Test)

JUNE 2019 SESSION

1 hour 30 minutes

Candidates answer on the question paper

Additional materials:

As listed in instructions to Supervisors

Calculator (optional)

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **both** questions.

Write your answers in the spaces provided on the question paper.

Use a sharp pencil for your drawings. Coloured pencils or crayons should not be used.

You should record all experimental results and show the essential steps in any calculation in the spaces provided in the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
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TOTAL	

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- 1** You are required to identify nutrients present in juice **X**.
 You are provided with three boiling tubes, a test tube rack, a pair of tongs, a graduated syringe, Benedict's solution, hot water bath, juice **X**, solution **P**, solution **R** and access to a clock.
- (a) (i) Carry out the tests described in Table 1.1 below and record the observations and deductions in the table.

Table 1.1

test	observations	conclusions
Add about 2 cm ³ of juice X into a boiling tube and add two drops of solution R to the juice.		
Place about 2 cm ³ of juice X into a clean boiling tube and add about 2 cm ³ of Benedict's solution to the juice. Heat the mixture in a water bath.		
Place about 3 cm ³ of juice X into a clean boiling tube and add about 4 cm ³ of solution P to the juice. Place the test tube in the hot water bath and leave it for about 3 minutes. After the 3 minutes, pour about half of the contents into another clean boiling tube. Add two drops of solution R to one of the portions.		
Add about 2 cm ³ of Benedict's solution to the other portion and heat in the water bath.		

[10]

(ii) Identify solution **R**.
..... [1]

(iii) State the colour of Benedict's solution.
..... [1]

(iv) State the nutrients contained in juice **X**.
.....
..... [2]

(v) Suggest **one** advantage of drinking juice **X**.
.....
..... [1]

(vi) Suggest a possible identity of solution **P**.
..... [1]

(vii) State the process responsible for the change that occurred when solution **P** was added to the juice and the mixture heated.
..... [1]

(viii) Name the part of the alimentary canal where the process mentioned in (vii) occurs.
..... [1]

(b) State any two precautions that should be taken when carrying out the experiment.
.....
.....
.....

- 2 (a) You are required to determine the resistance of a wire, **R**. **Fig.2.1** shows the circuit diagram for the circuit which the supervisor set for you.

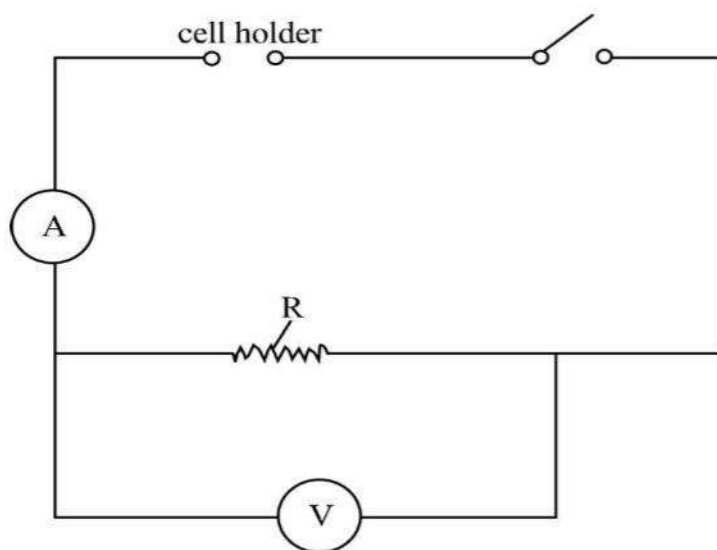


Fig.2.1

- (i) Place one cell in the cell holder, close the switch and record the voltmeter and ammeter readings in **Table 2.1**.

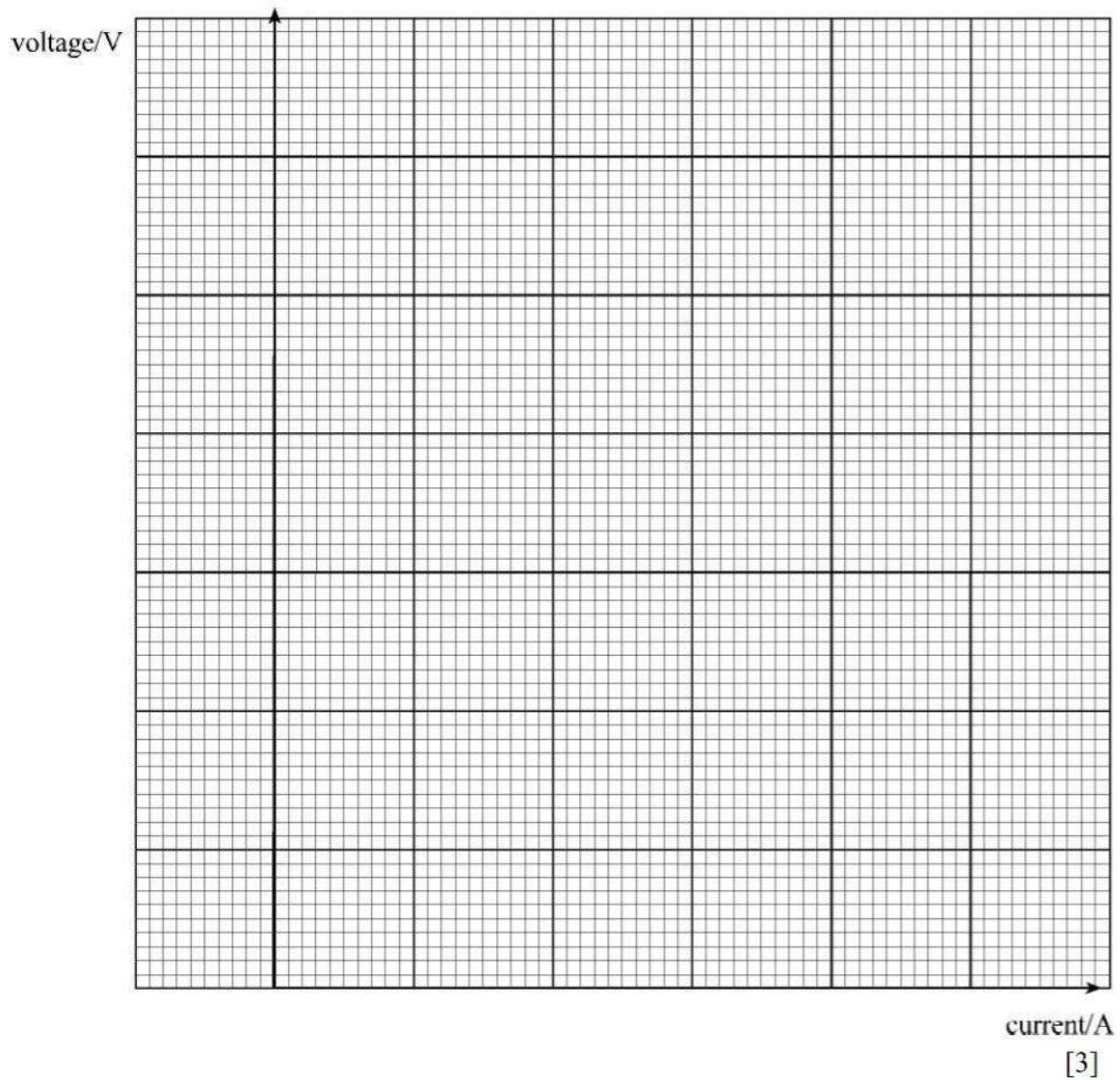
Repeat the procedure adding one cell at a time until you use four cells.

Table 2.1

number of cells	voltage/V	current/A

[10]

- (ii) Plot a graph of voltage (y-axis) against current (x-axis).



- (iii) State the relationship between voltage and current.

.....

..... [1]

(iv) Calculate, using the graph, the resistance of wire **R**.

[2]

(v) Determine, clearly showing on the graph, the current when the voltage is 3.5 V.

[2]

(b) State any two sources of error in the experiment.

.....
.....
.....

[2]